

Machine Learning

Consolidated Practice Question Bank

Assembled from ten past papers 2016–2021

About this document. Every question and solution from the ten solution PDFs in `PracticeProblems` appears exactly once below. Seventeen questions that were asked twice (or three times) across the papers have been merged; where the two source keys answered differently, both answers are kept. The grey line under each question number gives its provenance. Point values are inherited from the source papers and do not sum to a meaningful total.

Solutions are currently HIDDEN. Toggle the `\printanswers` line in the preamble.

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Part I Convexity and Optimization

I.1. (10 points) [QUIZ2 Q1 $\equiv$ END2018 Q5(a)]

(10=3+3+4) Define a convex set and a convex function mathematically. What is a convex optimization problem?

I.2. [QUIZ3 Q3 $\equiv$ END2018 Q5(a)]

(Extra Credit) Building upon the previous question, show that sum of any two convex functions is convex. Extend your argument to show that sum of any number of convex functions is convex.

I.3. (3 points) [END2018 Q5(b)]

Show that the loss functions for linear regression, ridge regression and LASSO are all convex functions with respect to the parameters.

I.4. [QUIZ4 Q1 $\equiv$ QUIZ3 Q1]

The binary cross-entropy loss is usually used with neural networks trained for a binary classification task. Write the mathematical form of the loss. Is it a convex function? Prove your claim.

I.5. (15 points) [QUIZ2 Q3]

Given the following optimization problem, answer the questions below

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_1^2 + x_2 \\ \text{subject to} \quad & 2x_1 + x_2 \geq 4 \\ & x_2 \geq 1 \end{aligned}$$

(a) (5 points) Show that this is a convex optimization problem.

(b) (4 points) Graphically indicate the *feasible region* of the solution within $\mathbb{R}^2$.

(c) (6 points) Write the Lagrangian for the above optimization problem. Make sure you specify the range of the Lagrange variables.

I.6. (10 points) [END2018 Q5(c)]

Using a simple (the simplest that you can come up with) neural network, show that learning the weights for a squared error loss term is a non-convex problem.

Part II Linear Regression and Regularization

II.1. (20 points) [MID2018 Q2]

Consider the following simple linear regression model.

$$y = wx + \epsilon$$

Where, y is the sum of deterministic linear function of x and some noise ϵ . x and y are the real-valued input and output respectively, w is the real-valued parameter to be learned. $\epsilon \sim N(0, \sigma)$ is a Gaussian random variable representing the noise.

- (a) (5 points) Write the expression for the probability distribution of y (i.e $p(y|w, x)$) in terms of $N(\cdot, \cdot), w, \sigma, x$.
- (b) (5 points) Suppose, we are given m i.i.d training samples $\{(y^1, x^1), \dots, (y^m, x^m)\}$, $\mathcal{Y} = \{y^1, \dots, y^m\}$ and $\mathcal{X} = \{x^1, \dots, x^m\}$. Derive an expression for conditional data likelihood (i.e $P(\mathcal{Y}|\mathcal{X}, w)$).
- (c) (5 points) Consider the conditional data likelihood, derive an expression for gradient descent learning rule.
- (d) (5 points) Suppose we are given a Laplacian prior ($p(x) = e^{-\lambda|x|}$, $x \in \mathbb{R}, \lambda > 0$) over w . What kind of regularization does it impose on your regular regression problem? Can you derive the expression for the log posterior? {Hint: Start with MLE, use Bayes rule and apply log, simple!}

II.2. (6 points) [END2018 Q1(c)]

The sum of squared error loss term for linear regression ($\mathbf{y} = \mathbf{X}\mathbf{w}$) given by the quadratic expression ($\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$), where each element of $\mathbf{y}$ and each row of $\mathbf{X}$ is one training sample pair. Starting with this loss, derive the closed form expression for the optimal parameters $\mathbf{w}^*$. Does ridge regression also have a closed form solution? If yes, what is it? If not, why not?

II.3. (4 points) [END2018 Q1(b)]

What are the pros and cons of ridge regression and LASSO?

II.4. (4 points) [MID2017 Q2(a) $\equiv$ END2019 Q5(a)]

What effect will adding a regularization term to a linear/logistic regression model have on the bias and variance of your model? Mention 'increase', 'decrease' or 'no change' along with a justification supporting your choice.

II.5. (3 points) [MID2016 Q2]

A locally weighted linear regressor outputs the mean of k nearest neighbors of a test point. Given the training data as $\{x_0 = 0.1, x_1 = 0.3, x_2 = 0.6, x_3 = 0.8\}$, determine the output of the regressor for $k = 2$ over the interval $0 \leq x \leq 1$.

II.6. (7 points) [END2017 Q6 $\equiv$ END2019 Q2(a)]

(7=2+1+1+2+1 points) Suppose you work for this to-become-great financial startup that gave you a lot of data for stock market prediction. Domain expertise and some quick verification of data led you to believe that the relationship between the input and the output variables are not linear. Unfortunately, you (or your more experienced colleagues) do not know the order of polynomial that will result in a good fit. However, you found that the data is smoothly varying. Given that you have limited computing resources (servers with large RAMs, but no GPUs) and are unable to train a deep neural network, your advisors have recommended using some variant of linear regression for quick prototyping. Succinctly answer the following questions for this scenario.

- a) What would be your modified loss function for the prediction problem?
- b) What would be the hyperparameters and how would you select them?
- c) How would you check if your model is correctly trained or not?

- d) Provide two different regularizers that you could use with these models? Also list one advantage each.
- e) Given the setup, what kind of an optimization technique would you chose? Provide a one line justification for your answer.

Hint: Smoothly varying data implies samples in a local neighborhood would have similar properties/structure. Do you know of any way this information could be leveraged?

II.7. (3 points) [MID2016 Q6]

(*Extra credit*) Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be i.i.d. data from a uniform distribution over the disc of radius θ in $\mathbb{R}^2$, i.e., $\mathbf{x}_i \in \mathbb{R}^2$ and

$$p(\mathbf{x}; \theta) = \begin{cases} \frac{1}{\pi \theta^2}, & \|\mathbf{x}\| \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2}$. What is the maximum likelihood estimate of θ .

II.8. (10 points) [MID2018 Q5]

{*Extra Credit*}: There is a heavy traffic jam at Connaught Place. There is bumper-to-bumper traffic and cars all around the outermost circle are at a standstill. Somehow Google leaks the GPS locations of all these cars. The typical noise in phone based GPS measurements would be around 10m. Google wants you to estimate the circumference of the outermost circle of Connaught Place (Trump asked them not to use other resources, but ask IIITD students to solve it). Since you have only done linear regression, you need to apply linear regression for this problem. First of all, can you apply *linear* regression? What would you need to do? Explain the entire training and validation pipeline.

Part III Logistic Regression and Classification Losses

III.1. (2 points) [MID2021 Q1(a)]

What is the loss function used to train a logistic regression model? Define the probabilistic model used, and write the mathematical expression for the loss (log-likelihood).

III.2. (2 points) [MID2021 Q1(b) $\equiv$ END2019 Q2(c)]

Write the regularized loss function for logistic regression with a Gaussian prior and a Laplacian prior (doubly exponential) on the weights. Which regularization function does each correspond to, and will the Laplacian prior have the same sparsity-inducing effect that it has in LASSO? Explain your answer.

III.3. (2 points) [MID2021 Q1(c) $\equiv$ END2019 Q4(c)]

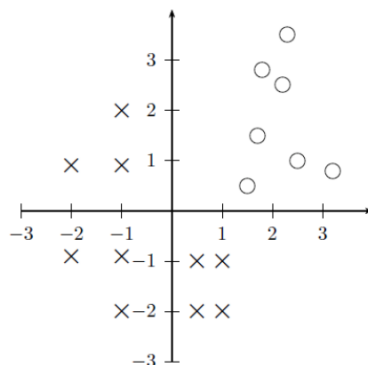
How would you extend or modify the binary classification loss to handle three (or more) different classes? What kind of activation would you have in the output layer of a neural network designed for multi-class classification?

III.4. (10 points) [END2018 Q1(a)]

Consider the regularized logistic regression loss function for the data shown in the figure

$$\max \sum_{i=1}^N \log \left(P(y^i | x_1^i, x_2^i) \right) - \lambda_0 w_0^2 + \lambda_1 w_1^2 + \lambda_2 w_2^2$$

where i indexes the i^{th} data point and $P(y|x_1, x_2) = \frac{1}{e^{w_0 + w_1 x_1 + w_2 x_2}}$. Describe what would be the learned weights (w_0, w_1, w_2) be in the following four scenarios. Instead of giving exact values of w_i , reason about what values would they take in the context of λ_i .



- (i) $\lambda_0 = \lambda_1 = \lambda_2 = 0$ (ii) $\lambda_0 = \infty; \lambda_1 = \lambda_2 = 0$
 (iii) $\lambda_0 = \lambda_1 = 0; \lambda_2 = \infty$ (iv) $\lambda_0 = \lambda_2 = 0; \lambda_1 = \infty$
 (v) $\lambda_0 = \lambda_1 = \lambda_2 = 1/3$

III.5. (5 points) [MID2016 Q5]

Consider the following training set of four labeled data:

x_1	x_2	y
1	2	1
3	2	1
2	2	0
0	3	0

The logistic regression model is: $p(y = 1|\mathbf{x} = [x_1 \ x_2]^\top) = \frac{1}{1 + e^{-\mathbf{x}^\top \boldsymbol{\theta}}}$, where $\mathbf{x}^\top = \theta_0 + \theta_1 x_1 + \theta_2 x_2$. Estimate the parameters θ_0 , θ_1 , and θ_2 .

III.6. (5 points) [END2019 Q2(b)]

You have points along the surface of two concentric spheres in 3D, each belonging to a different class. How would you apply logistic regression to classify these samples?

Part IV Naive Bayes and Generative Classifiers

IV.1. (5 points) [MID2018 Q4(a)]

Explain why the most probable label for the training sample $(x_1, x_2, \dots, x_m)$ is the label l that maximizes the following:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_m = x_m | L = l) P(L = l)$$

IV.2. (5 points) [MID2017 Q3(b) $\equiv$ MID2018 Q4(b)]

To build a spam/no-spam classifier that uses text frequency features and uses a small training set, one will need to consider the scenario where some features have rare values that were *never* observed in the training data (e.g., the word ‘Thiruvananthapuram’ does not appear in training for an email classification problem). Assume that a word is never observed in the training set, but is present in a test email sample. Does such a situation pose a challenge for the Naive Bayes Classifier? If yes, how would you fix this issue? If not, then what strategy helps prevent this issue in NB?

IV.3. (6 points) [END2017 Q4]

Consider the Naive Bayes classifier trained on the following dataset:

x_1	x_2	x_3	y
1	1	1	1
1	0	1	1
1	1	0	1
0	0	1	0
0	0	0	0

What is the prediction on y for the input sample's feature $x_1 = 1, x_2 = 0, x_3 = 0$?

IV.4. (8 points) [END2017 Q3]

(8=2+3+3 points) We are given a two class dataset with class labels $y \in \{0, 1\}$ described by a single feature x . The dataset has equal number of samples of each class. Say, we decide to build a binary Gaussian Naive Bayes (GNB) Classifier.

- (2 points) Write the expression of the decision rule.
- (3 points) We are given that the feature of the two classes have unequal means ($\mu_1 = 0$ and $\mu_2 = 2$) but equal variances ($\sigma_1^2 = 1$). Is the decision boundary linear? What is the expression of the decision boundary? Justify your answer. Draw the graph showing the distribution under the two classes along with the decision boundary.
- (3 points) We are given that the feature of the two classes have equal means ($\mu_1 = \mu_2 = 0$) but unequal variances ($\sigma_1^2 = 0.25$ and $\sigma_2^2 = 1$). What is the expression of decision boundary? Justify your answer. Draw the graph showing the distribution under the two classes along with the decision boundary.

IV.5. (2 points) [MID2017 Q3(a)]

In online learning, we can update the decision boundary of a classifier based on new data without reprocessing the old data. Now for a new data point that is an *outlier*, which of the following classifiers are likely to be affected more severely? Naive Bayes (NB), Logistic Regression (LR), Support Vector Machine (SVM)? **{Note: An outlier is a point with lies in a low-probability region, typically farther away from most points of that class, e.g., for a gaussian random variable $\mathcal{N}(\mu, \sigma^2)$, an outlier would be a point that lies beyond the $\mu \pm 3\sigma$ range.}**

IV.6. (2 points) [MID2017 Q3(c)]

Now to build a classifier on high-dimensional features using small training data, one will need to consider the scenario where many features are just irrelevant noise. To train a generalizable classifier, would you want to use NB or LR? How will you augment the original formulation of the classifier under a Bayesian or regularization setting, i.e., which regularizer or prior will you incorporate in the likelihood term?

Part V Nearest Neighbours and Instance-Based Methods

V.1. (2 points) [MID2017 Q2(b)]

What effect will increasing k in k -nearest neighbor models have on the bias and variance of your model? Mention ‘increase’, ‘decrease’ or ‘no change’ along with a justification supporting your choice.

V.2. (6 points) [MID2018 Q4(c)]

True or False ?. Justify your answer.

- (i) In a nearest neighbor classifier, Euclidean distance and squared Euclidean distance are equivalent.
- (ii) The 3-nearest neighbor classifier is always more accurate than the 2-nearest neighbor classifier.
- (iii) With sufficient training data, the error of a nearest neighbor classifier always goes down to zero.

V.3. (2 points) [MID2018 Q4(d)]

Give scenarios where you would prefer to use k -nearest neighbors instead of Support Vector Machines.

V.4. (2 points) [MID2018 Q4(e)]

k -nearest neighbors is a nonparametric classifier because you need to retain all the training samples in order for it to work well. If I say SVMs are not nonparametric, would I be correct or wrong? Justify your answer.

Part VI Support Vector Machines and Kernels

VI.1. (3 points) [MID2017 Q4 $\equiv$ QUIZ3 Q2]

- (a) Write down the soft-margin primal formulation of SVM. Let ‘ C ’ denote the regularization parameter that weights the classification loss term (not $w^\top w$). In the figure below, the solid lines denote the decision boundary, the *solid* squares (circles) indicate the support vectors and the squares (circles) denote the points in the input space. The two images are a result of using different kernel and different regularization parameters. Which of the following configurations of SVM would describe the images best? List all that apply to the figures (a) and (b), with one-line justification.
- a) soft-margin SVM with small C
 - b) soft-margin SVM with large C
 - c) hard-margin SVM with polynomial kernel, i.e., $k(x, y) = (x^\top y)^p$, $p > 1$
 - d) hard-margin SVM with radial basis function kernel, i.e., $k(x, y) = \exp(-\gamma \|x - y\|_2^2)$, $\gamma > 0$

- e) none of the above.
- (b) The primal formulation for the hard-margin SVM is convex. Prove that the soft-margin objective is still convex.

VI.2. (10 points) [QUIZ2 Q2 $\equiv$ END2018 Q2(a)(i); END2018 Q2(a)(ii,iii)]

- (a) Suppose you have identified the maximum margin separating hyperplane $\mathbf{w}^*$. Given points $\{\mathbf{x}_i^+\}$, $i = 1, 2, \dots, M$ and $\{\mathbf{x}_i^-\}$, $i = 1, 2, \dots, N$, that you want to classify, how would you compute the optimal offset b^* ?
- (b) Consider the training data samples and the corresponding Lagrange multipliers learned from them, as given in the following table. Identify the support vectors.

i	x_i	y_i	λ_i	i	x_i	y_i	λ_i
1	(4,2.9)	1	0.414	6	(1.9,1.9)	-1	0
2	(4,4)	1	0	7	(3.5, 4)	1	0.018
3	(1,2.5)	-1	0	8	(0.5,1.5)	-1	0
4	(2.5,1)	-1	1.18	9	(2,2.1)	-1	0.414
5	(4.9,2.5)	1	0	10	(4.5,2.5)	1	0

- (c) Compute w and classify the point (3,3).

VI.3. (5 points) [MID2016 Q4]

Consider the training dataset $\{(x_1, y_1) = (-2, -1), (x_2, y_2) = (-0.5, -1), (x_3, y_3) = (0.5, 1), (x_4, y_4) = (2, 1)\}$, where x_i are the data points and y_i are the corresponding label. If you chose to learn an SVM using this training data, how would you formulate the primal problem? What is the width of the margin in the input space? If we use a feature mapping $\phi(x_i) = [x_i \ x_i^2]^\top$ with this dataset, followed by learning a linear SVM in the new feature space, what would be the width of the margin? In general, if we use a mapping $\psi(x_i) = [x_i, x_i^2, \dots, x_i^n]$ for this data, how would the SVM margin change?

VI.4. (2 points) [MID2017 Q2(d)]

What effect will removing all the non-support vectors in SVMs have on the bias and variance of your model? Mention ‘increase’, ‘decrease’ or ‘no change’ along with a justification supporting your choice.

VI.5. (4 points) [END2017 Q2(a)]

What is the leave-one-out cross-validation error estimate for maximum margin separation in Figure-1 below?



Figure 1: Sample Space

VI.6. (8 points) [END2017 Q2(b)]

Given a standard linear binary SVM classifier.

$$\min \frac{\mathbf{w}^\top \mathbf{w}}{2}, \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \quad i \in 1, 2, \dots, l. \quad (1)$$

Can we increase the physical margin width by defining

$$\min \frac{\mathbf{w}'^\top \mathbf{w}'}{2}, \quad \text{s.t.} \quad y_i(\mathbf{w}'^\top \mathbf{x}_i + b') \geq 2 \quad i \in 1, 2, \dots, l? \quad (2)$$

Justify your answer with mathematical reasoning.

VI.7. (8 points) [MID2017 Q5 $\equiv$ END2018 Q2(c)]

Given n training examples $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$, let elements of the $n \times n$ kernel matrix be $\mathbf{K}_{ij} = \Phi(\mathbf{x}_i)^\top \Phi(\mathbf{x}_j)$.

- (a) (3 points) Prove that the kernel matrix $\mathbf{K}$ is positive semidefinite. *Hint:* Recall that an $n \times n$ matrix $\mathbf{A}$ is positive semidefinite if and only if for all $\mathbf{y} \in \mathbb{R}^n$, we have $\mathbf{y}^\top \mathbf{A} \mathbf{y} \geq 0$.
- (b) (5 points) Mercer's theorem tells us that $K(x; z)$ is a valid kernel function if and only if for any finite set $\{x_1, \dots, x_m\}$, the square matrix $K \in \mathbb{R}^{m \times m}$ is symmetric and positive semidefinite, where the entries of the matrix are given by $K_{ij} = K(x_i; x_j)$. Recall that a positive definite matrix K satisfies $x^\top K x > 0$, $\forall x \in \mathbb{R}^m, x \neq 0$. K also has only positive eigenvalues, and every submatrix containing the diagonal elements of K , i.e., every submatrix of size $k \times k$, $k = 1, 2, \dots, m$ starting from the top left entry has a positive determinant.

Let K_1, K_2 be kernels over $\mathbb{R}^n \times \mathbb{R}^n$, let $a \in \mathbb{R}^+$ be a positive real number. For each of the functions K below, state whether it is necessarily a kernel. If you think it is, prove it; if you think it isn't, give a counter-example.

- (i) $K(x, z) = K_1(x, z) + K_2(x, z)$
- (ii) $K(x, z) = K_1(x, z) - K_2(x, z)$
- (iii) $K(x, z) = aK_1(x, z)$
- (iv) $K(x, z) = -aK_1(x, z)$
- (v) $K(x, z) = K_1(x, z)K_2(x, z)$

VI.8. (4 points) [END2019 Q3(a)]

(2+2) The ξ^2 kernel is defined as $\kappa(x, y) = \sum_{i=1}^D \frac{x_i y_i}{x_i + y_i}$, where $x_i, y_i \geq 0$ and $\sum_{i=1}^D x_i = \sum_{i=1}^D y_i = 1$. Text data is often represented using feature vectors that are like histograms, frequently referred to as *bag of words*. Explain how would you use ξ^2 kernel to compute distances between two such feature points. Why is it justified to use this specific kernel? {*Hint:* Recall that kernel functions represent inner products between vectors in some feature space. How can you use inner products to compute distances?}

VI.9. (8 points) [END2019 Q3(b)]

(2+6) What is complementary slackness with respect to a convex optimization problem? Does it apply to support vector Maxine Learnings? Explain the implications of the complementary slackness condition by writing all the constraints on the primal and dual variables for the SVMs.

VI.10. (4 points) [END2019 Q3(c)]

Show that the constraints in a primal SVM formulation are affine, i.e., simultaneously convex and concave.

VI.11. (4 points) [END2019 Q3(d)]

How does the Lagrangian help in converting a constrained optimization problem to an unconstrained optimization problem. Starting from the original SVM formulation as a constrained optimization problem, develop the Lagrangian, and then the primal problem. Explain the equivalence of the primal problem with the original SVM formulation with constraints.

VI.12. (5 points) [MID2017 Q6]

(Extra credit)

- Determine the mapping $\phi(x)$ such that the kernel $k(\mathbf{x}, \mathbf{z}) = (c + \mathbf{x}^\top \mathbf{z})^2 = \phi(\mathbf{x})^\top \phi(\mathbf{z})$, where $\mathbf{x} = (x_1, x_2)^\top$ and $\mathbf{z} = (z_1, z_2)^\top$.
- Show that an XOR is not linearly separable, but that after the mapping $\phi(\mathbf{x})$ with $c = 0$, it is linearly separable.

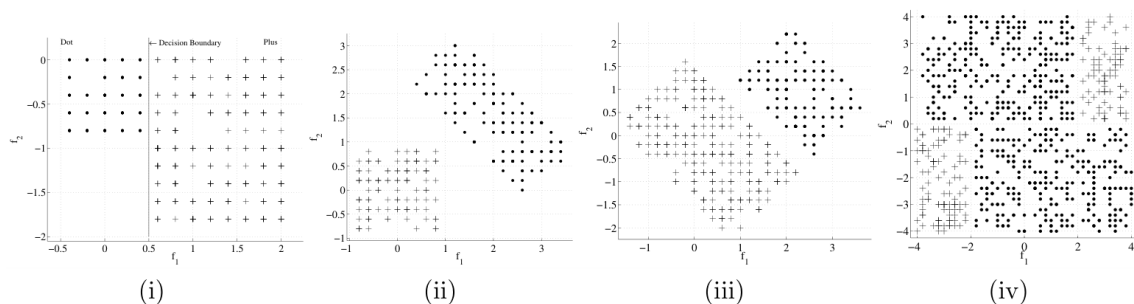
Part VII Decision Trees and Ensembles

VII.1. (5 points) [MID2017 Q1]

You are given points from 2 classes, shown as +’s and ·’s. For each of the following sets,

- Draw the decision tree of depth at most 2 that can separate the given data completely, where each node involves thresholding of a *single* variable.
- Draw the decision boundaries corresponding to your decision tree in your answer sheet, clearly showing the axes and slope/intercept. Write the class labels for each of the resulting regions.

If in any of the cases, the data can not be separated completely by a depth 2 decision tree, can the data be separated using a decision tree of higher depth? If yes, can you draw one?



VII.2. (2 points) [MID2017 Q2(c)]

What effect will pruning a decision tree (to a certain depth) have on the bias and variance of your model? Mention ‘increase’, ‘decrease’ or ‘no change’ along with a justification supporting your choice.

VII.3. (20 points) [END2018 Q4]

Consider the training dataset given below. In the dataset, X_1 , X_2 , and X_3 are the attributes and Y is the class variable.

Sample	X_1	X_2	X_3	Y
S1	0	0	0	+
S2	0	0	1	-
S3	0	1	0	-
S4	0	1	1	+
S5	1	0	0	-

Sample	X_1	X_2	X_3	Y
S6	1	1	0	+
S7	0	1	1	-
S8	1	1	1	-

Table 1: Left: Training set; Right: Validation set

- (a) (5 points) Which attribute has the highest information gain? Justify your answer?
- (b) (9 points) Draw the (full) decision tree for this dataset using the information gain criteria.
- (c) (6 points) Can you construct a decision tree that will have 100% accuracy on the validation dataset as well as the training dataset. Answer Yes or No. If your answer is yes, then draw the decision tree that will have 100% accuracy. If your answer is no, then explain why it cannot be done. No credit will be given if the explanation is incorrect.

VII.4. (5 points) *[END2019 Q1(b)]*

(1+2+2) What is entropy? Give the mathematical definition of entropy. Justify or refute the statement “A node should be a leaf node if its entropy is zero”. Can you give one (or more) scenarios where you may choose to have a leaf node that has non-zero entropy?

VII.5. (5 points) *[END2019 Q1(a)]*

We have seen decision trees for classification in class. How would you extend them to perform a regression task? What would be the advantage of doing so over linear regression?

VII.6. (5 points) *[END2019 Q1(c)]*

Give a total of five advantages and disadvantages of decision trees over classifiers like SVMs or logistic regression. (1 mark for each point)

VII.7. (4 points) *[END2017 Q1]*

Why are “weak” learners used with boosting method?

Part VIII Neural Networks and Deep Learning

VIII.1. (6 points) *[END2017 Q5(a) $\equiv$ END2019 Q4(b) $\equiv$ END2019 Q4(f), rst half]*

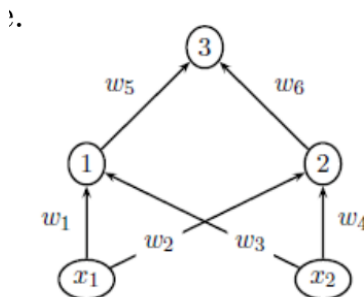
(2+2+2 points) Write the mathematical expressions of sigmoid, tan-hyperbolic, and linear rectified activation functions used in neural networks. State the pros and cons of these activation functions. Compute the derivative of each, explain why ReLU is expected to work better in deeper networks than sigmoid or tanh, and say whether a tanh activation can be used at the output layer of a binary classifier.

VIII.2. (5 points) *[END2019 Q4(a)]*

Give the expression for binary cross-entropy (BCE) loss. Usually, it is used with a sigmoid output activation. Argue why is the BCE loss more effective than the mean-squared error (MSE) loss. Make use of corresponding gradients to support your arguments.

VIII.3. (10 points) [END2018 Q3(b)]

Consider the neural network shown in the figure below. Assume that all internal nodes and output nodes compute the $\tanh()$ activation function. Using the squared error loss, derive the weight update equations using backpropagation for $(w_1, \dots, w_6)$. {Hint: $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$; Try simplifying the gradient of $\tanh()$ as you would do for sigmoid.}



VIII.4. (6 points) [END2017 Q5(b)]

(2+2+2 points) Given a neural network in Figure-2 below:

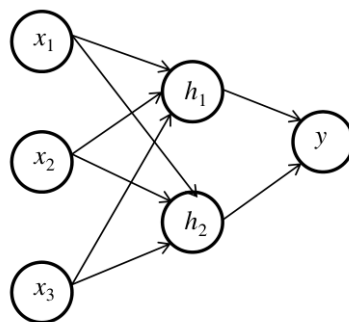


Figure 2: Neural Network

We use linear rectified activation function σ at the hidden layer and output layer neurons. The loss function is given by $L = \frac{1}{2}(y - t)^2$, where t denotes the target value. Let us denote the hidden layer and output layer weights by matrices $\mathbf{W}$ and $\mathbf{V}$, respectively, that are initialized as below:

$$\mathbf{W} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{V} = \begin{bmatrix} 0 & 1 \end{bmatrix}. \quad (1)$$

- Write the expression of mapping x to y in terms of σ , $\mathbf{W}$, and $\mathbf{V}$.
- Assume that the current input is $\mathbf{x} = [1 \ 2 \ 1]$. The target value is $t = 1$. Compute the numerical output value y . Show all intermediate steps.
- Compute the gradient of the loss function with respect to the weights $\mathbf{V}$ and $\mathbf{W}$. Also, compute the values numerically for the choices of $\mathbf{W}$, $\mathbf{V}$, $\mathbf{x}$, and t given above.

VIII.5. [QUIZ3 Q4]

Consider a neural network with only one hidden layer comprising of two neurons, each having relu activations ($\text{relu}(z) = \max(0, z)$). Assume that the neural network is performing a binary classification, implying the output layer has a single neuron with a sigmoid activation. Draw this neural network for two dimensional input data. Derive the gradient update rule for each of the weights using backpropagation.

VIII.6. (10 points) [END2018 Q3(a) $\equiv$ QUIZ4 Q3]

What are the different components of a *deep* Convolutional Neural Network applied to classification? Describe each component based on its need, functionality and advantages over traditional neural network architectures (multi-layer perceptrons).

VIII.7. [QUIZ4 Q4]

Why are the convolution and pooling layers alternately placed in a typical CNN architecture? Could you achieve a similar effect as pooling by changing a hyperparameter like stride?

VIII.8. (5 points) [END2019 Q4(d)]

The first layer of AlexNet has 96 filters of size 11×11 , that process an input image of $224 \times 224 \times 3$ to generate feature maps of size 55×55 . What is the stride used in the first layer? The last max pooling layer of size $(13 \times 13 \times 256)$ is connected to a two layered fully-connected layer of size 4096, followed by a 1000-node softmax based output layer. What is the total number of parameters for the last four layers (max pooling, linear, linear and softmax)?

VIII.9. (4 points) [END2019 Q4(f), second half $\equiv$ END2019 Q5(b)]

What is dropout, and why is it effective in reducing overfitting? What other strategies would you use to avoid overfitting in deep neural networks?

Part IX Unsupervised Learning and Dimensionality Reduction

IX.1. (2 points) [MID2016 Q1]

What is the objective function of k -means clustering? How would you choose an appropriate value of k ?

IX.2. (5 points) [MID2016 Q3]

Write the expression for the pdf arising from a K mixture of Gaussians. Given an i.i.d. sample of N data points, what will be the log likelihood function that you would maximize to estimate the parameters? Derive the expression for the mixing coefficients π_k . If the k^{th} Gaussian tends to collapse to a single point ($\pi_k \rightarrow 1/N$, $\boldsymbol{\mu}_k \rightarrow \mathbf{x}_i$, $\Sigma_k \rightarrow \mathbf{0I}$), will the likelihood function value tend to increase? Explain why or why not. Given your goal is to model the data using a mixture of Gaussians, comment on what needs to be done, if anything at all.

IX.3. [QUIZ4 Q2]

Principal Component Analysis (PCA) is powerful tool for dimensionality reduction. It finds a linear mapping $x_{low} = Wx_{high}$ such that the inverse mapping $\hat{x}_{high} = W'x_{low}$ has a small ℓ_2 error with respect to the original x_{high} . The lower-dimensional representation is often used for ML tasks like classification, etc. Can you design the simplest neural network that is capable of implementing PCA exactly? Give the architecture (number of layers, activation, etc.) and the loss function that you would use.

Part X Model Evaluation, Bias–Variance and Cross-Validation

X.1. (4 points) [MID2018 Q3(a)]

Write mathematical formulation of one performance evaluation metric for each of the following tasks.

(i) Classification, (ii) Regression

X.2. (4 points) [MID2018 Q3(b)]

What are the drawbacks of holdout set? Also, suggest an alternate method.

X.3. (4 points) [MID2018 Q3(c)]

How is K-fold cross validation different from Leave-one-out Cross validation? (write any two differences). If the data is sparse, which cross-validation method would be beneficial?.

X.4. (4 points) [MID2018 Q3(d)]

Write two pros and cons of choosing (i) Large number of folds, (ii) Small number of folds in a cross-validation task:

X.5. (4 points) [MID2018 Q3(e) $\equiv$ END2019 Q5(c)]

Explain how would you detect (i) High variance, (ii) High bias, in your learning model. Write one solution each for fixing them.

X.6. (1 point) [MID2021 Q2(a) $\equiv$ END2018 Q2(b)]

You are given a data set for detecting a rare type of cancer. You've build a classification model and achieved a validation accuracy of 96%. Why shouldn't you be happy with your model performance? Is accuracy the right metric for this problem? Justify your argument {Hint: What is the fraction of people who have this type of cancer as opposed to people who do not?}

X.7. (1 point) [MID2021 Q2(b)]

You are working on a classification problem. For validation purposes, you've randomly partitioned the training data into a train subset and a validation subset. You are confident that your model will work incredibly well on unseen data since your validation subset accuracy is high. However, you get shocked after getting poor test accuracy. What could have gone wrong? How would you resolve it?

X.8. (2 points) [MID2021 Q2(c)]

Assume that you have a large dev set on which your classifier achieves a 20% error. Describe using examples, how would you use *error analysis* to improve your classifier's performance. Hint: Is there a way to discover potential sources of error?

Part XI Sequential Decision Making

XI.1. (6 points) [END2017 Q7]

(6=1+1+1+1+2 points) Long after winter has come and gone, the world of Game of Thrones has transitioned from barbaric to capitalistic. Gilly's son, Sam, is growing up north of the wall and has left home with a 100 pounds, seeking his own adventures. The whitewalkers (WW) led by the night king lost the war and are managing survival by stealing money from young kids. Sam has a 20% chance of running into WW. If the WW find Sam, he loses all his money. Let's assume that Sam's reward function is $R(x) = \log(1 + x)$. If WW find him, his utility is $R(0) = \log(1 + 0) = 0$, and if they don't, then it is $R(100) = \log(1 + 100)$.

- If you have no control over Sam's actions, what is the expected reward for Sam?
- LittleFinger offers theft insurance to Sam: If Sam pays 30 pounds, then LittleFinger reimburses the entire 70 pounds in case of theft (note that Sam has paid 30 pounds as premium already, so only has a balance of 70 pounds). What is the expected reward for Sam if he takes this insurance? If the goal is to maximize the reward, should Sam take the insurance?
- In the previous scenario, from LittleFinger's point of view, does it make sense to sell this insurance? What is the expected profit that LittleFinger would make?
- If Sam were to take actions through a deterministic policy, the value function for the policy would be

$$V^\pi(s) = \sum_{s'} T(s, \pi(s), s') R(s, \pi(s), s') + \gamma V^\pi(s')$$

Define stochastic policy and write the expression for value function under such a policy.

- Since Sam is new to RL, affirm or refute his argument of "If the only difference between two MDPs is the discount factor γ , then they must have the same policy" with a justification or counter-example.

Note: some numbers that you may need $\log(101) \simeq 4.6151$, $\log(71) \simeq 4.2627$.

Part XII True/False and Multiple-Choice Blocks

XII.1. (20 points) [MID2018 Q1]

Answer the following True/False questions. An answer is true if it is *always* true. You have to justify your answer to get full credit.

- (2 points) True or False ?
Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ be the two arbitrary matrices.
(i) $\text{rank}(AB) \geq \min(\text{rank}(A), \text{rank}(B))$. (ii) if $m = n$, $\text{rank}(A) = \text{rank}(A^{-1})$.
- (2 points) As number of samples increases, the MAP and MLE estimate become the same.
- (2 points) Naive Bayes can only classify linearly separable data.
- (2 points) Naive Bayes can handle only discrete valued variables.
- (2 points) In logistic regression, there exists a closed form solution for the parameters that maximizes the conditional log likelihood.
- (2 points) Given a linearly separable data, the margin of the decision boundary produced by SVM will always be greater than or equal to the margin of the decision boundary produced by any other hyperplane that perfectly classifies that data.

- (g) (2 points) If my training error is too high, I should collect more training data to resolve this problem.
- (h) (2 points) If my model is overfitting, I should collect more data to fix it.
- (i) (2 points) If my features/observations/input variables are all independent, it would not matter whether I use a Bayes Classifier or a Naive Bayes classifier.
- (j) (2 points) Normalizing data is for kids. Real experts work with raw data!

XII.2. (3 points) [MID2021 Q3]

Answer True or False with justification.

- (a) A test set is redundant if you have a sufficiently large dev set.
- (b) A logistic regression model uses the logit function to handle nonlinear boundaries in the data.
- (c) Gradient descent based approaches are useless for linear regression problems due to the existence of a closed form solution.

XII.3. (2 points) [MID2021 Q4]

Answer the following questions with justification.

- (a) k -fold cross-validation is useful for the following.
 - A. Hyperparameter tuning
 - B. Comparing classifiers
 - C. Determining underfitting and / or overfitting
 - D. All of the above
 - E. None of the above
- (b) A receiver receives two types of signals, zero and non-zero. The zero signal is distributed as $p(v|y = y_z) \sim \mathcal{N}(0, 4)$, i.e., a Gaussian distribution with mean $\mu = 0$ and variance $\sigma^2 = 4$. The non-zero signal generates a voltage distributed as $p(v|y = y_{nz}) \sim 0.5\mathcal{N}(-5, 4) + 0.5\mathcal{N}(+5, 4)$. Your task is to design a classifier that, given the voltage, can identify whether the signal is zero or non-zero. Please select all that are correct from the following choices with appropriate justification.
 - A. A Naive Bayes classifier will serve as a good solution.
 - B. Logistic regression may not work, but a Bayesian logistic regression (i.e., logistic regression with an appropriate regularization) will work as a good solution.
 - C. All of the above
 - D. None of the above